

Probabilistic load forecasting for buildings considering weather forecasting uncertainty and uncertain peak load

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HIGHLIGHTS

- A probabilistic load forecasting model for buildings is developed.
- Weather forecasting uncertainty is quantified and used for load forecasting.
- Uncertain peak load is modeled considering its magnitude and occurrence.
- A hybrid method is proposed to decompose the original load dataset.
- Case studies are conducted to compare different means of using weather forecasts.

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ABSTRACT

This paper presents a new probabilistic building load forecasting model considering uncertainties in weather forecasts and abnormal peak load. Two basis function components are developed, including the probabilistic normal load forecasting and the probabilistic uncertain peak load (or abnormal peak load) forecasting. The probabilistic normal load forecasting model is built using the artificial neural network (ANN) and the probabilistic temperature forecasts. The probabilistic abnormal peak load forecasting model consists of two models quantifying the probabilistic occurrence and magnitude of the peak abnormal differential load respectively. The test results show that the ANN deterministic load forecasting model can achieve satisfactory performance with the average mean absolute percentage error (MAPE) of 5.0%. The probabilistic occurrence model can forecast the occurrence frequency of the peak abnormal differential load with the satisfactory agreement, and the probabilistic magnitude model can well forecast the magnitudes of the peak abnormal differential load with the Kolmogorov-Smirnov error of 0.09. Real-time application case studies are conducted by different means of using the probabilistic weather forecasts. The results show that the probabilistic normal load forecasts have satisfactory accuracies and the load forecasts based on the one-day-ahead probabilistic weather forecasts are the best.

1. Introduction

Electric load forecasting has been playing a significant role in the electric power industry for over a century [1]. The conventional means of load forecasting is deterministic electric load forecasting, which has been widely applied in the control of power plants and electric power exchange in interconnected systems [2]. However, modern upgrades of power grids have resulted in more fluctuations of the electricity generations in the supply side, which exerts more pressure on the demand-side management (DSM). Additionally, various uncertainties of building use and working conditions as well as extreme events often lead to significant fluctuations of electricity load in the demand side,

which leads to the need of the peak load management (or demand limiting) [3]. Hence, given uncertainties from both the grid supply and demand sides, probabilistic load forecasting has attracted increasing attention due to its capability of providing more comprehensive information in the decision-making process [4].

Conventional deterministic electric load forecasting is usually grouped into three categories based on its forecast horizon: short-term load forecast (for one day to one week), medium-term load forecast (for one week to one month), and long-term load forecast (for more than one month). Most recent studies on deterministic load forecasting utilize the computational techniques including support-vector machines [5], artificial neural network (ANN) [6], expert systems [7], and fuzzy

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Nomenclature

PAD	peak abnormal differential	$P_{PAD,pre}^{m,h}$	quantile at the hour h (kVA)
ANN	artificial neural network	$P_{PAD}^{m,h}$	predicted probability of the PAD load occurrence at the hour h and the month m
LM	Levenberg-Marquardt		empirical probability of the PAD load occurrence based on the historical data
ECDF	empirical cumulative distribution function	f	probability density function of a variable
HKO	Hong Kong Observatory	μ	mean parameter of Normal distribution
PDF	probability density distribution	σ	standard deviation parameter of Normal distribution
K-S	Kolmogorov-Smirnov	α	shape parameter of Gamma distribution
LR	linear regression	β	inverse scale parameter of Gamma distribution
LOF	local outlier factor	Γ	complete Gamma function
KDE	kernel density estimation	λ	scale parameter of Weibull distribution
CDF	Cumulative distribution function	k	shape parameter of Weibull distribution
MAPE	mean absolute percentage error	$\emptyset$	mean parameter of $\ln(x)$ in Lognormal distribution
MAE	mean absolute error	γ	standard deviation parameter of $\ln(x)$ in Lognormal distribution
RMSE	root mean squared error	$\mathbf{z}^a$	actual target output vector
mse	mean square error	$\mathbf{z}$	network output vector
T_{min}	probabilistic daily minimum temperature forecast (°C)	C	calculated cumulative distribution function for the peak abnormal differential load based on the fitted probability density function
F_{Tmin}	forecasted probability of the daily minimum temperature with a certain value	O	observed cumulative distribution function based on the actual peak abnormal differential load data
P	probability of a variable	$L_{gap,t}$	gap in the original time series load dataset at the time t
T_h	hourly deterministic temperature forecast at the hour h (°C)	L_t	estimated hourly load (kVA)
T^{max}	maximum value of all hourly deterministic temperature forecasts in a day (°C)	$\hat{f}_{hbw}$	kernel density estimation function
$T_{h,n\%}$	hourly probabilistic temperature forecast at the $n\%$ quantile at the hour h (°C)	K	kernel smoothing function
$T_{max,n\%}$	daily probabilistic maximum temperature forecast at the $n\%$ quantile (°C)	h_{bw}	bandwidth of the Kernel density estimation function
T^{min}	minimum value of all hourly deterministic temperature forecasts in a day (°C)	LOF	degree of an object as a local outlier
$T_{min,n\%}$	daily probabilistic minimum temperature forecast at the $n\%$ quantile (°C)	lrd_k	local density of an object in its k -distance neighborhood
$L_{h,n\%}$	hourly probabilistic normal load forecast at the $n\%$ quantile at the hour h (kVA)	N_k	k -distance neighborhood
		$\hat{F}_n$	empirical cumulative distribution function
		$y_{noz,i}$	normalized value of the variable y

logic [8]. Compared to other computational techniques, the ANN shows superior performance due to its better capability of mapping the relationship between the inputs and output(s) without making complex mathematical formulations [9].

Compared to the deterministic load forecasting, the probabilistic load forecasting is presented in the form of density, quantiles, or intervals [4]. Recently, a tutorial review has summarized the significant development in the probabilistic electric load forecasting, including notable techniques, methodologies and evaluation methods [4]. It also concludes that three aspects or any two or three of them can be used to generate the probabilistic load forecasts:

- Simulating input scenarios: An integrated probabilistic electric load forecasting is proposed, which involves the forecast combination of deterministic load forecasting models and temperature scenarios based probabilistic forecasting [10].
- Use of probabilistic models (e.g., quantile regression): The probabilistic load forecasts are generated by performing quantile regression averaging on a set of sister deterministic forecasts and the proposed approach leads to dominantly better performance measured by the pinball loss function [11].
- Converting deterministic forecasts into probabilistic forecasts by residual simulation: A model is presented for probabilistic forecasts of both magnitude and timing [12].

However, most existing studies on the probabilistic load forecasting for buildings implement temperature scenarios generated from historical climate data, instead of the real probabilistic temperature forecasts, which do not make use of updated online information on future

weather condition and cannot provide the best prediction of the building load. In addition, most of these studies are conducted on an aggregation level or system-level loads [13]. There are very few studies on probabilistic load forecasting for individual load profiles, which have large uncertainties and therefore difficult to be predicted. Recently, a probabilistic individual load forecasting using pinball loss guided LSTM has been conducted for both residential and commercial buildings [13]. Also, the probabilistic forecasting of residential electricity consumption of a single household is conducted using the Gaussian Processes [14]. However, these models are only for short-term probabilistic load forecasting or a residential building and neglect the uncertainty of peak load. As for individual commercial/non-residential buildings, the individual load uncertainty significantly depends on the uncertain peak load and the weather forecasting uncertainty (for that the HVAC load is highly related to weather conditions).

As aforementioned, the uncertain peak load in individual buildings is a significant issue for probabilistic load forecasting. The literature on the building peak load forecasting is quite limited. Most of the previous studies on the peak load forecasting try to construct deterministic models on related information such as weather and historical load data. In 2001, the EUNITE competition was organized aiming to predict the daily maximum load of the next 31 days and several computational techniques (e.g., the support vector machine [15], the hybrid artificial intelligence scheme [16] and ANN) are used. However, the standard datasets provided in the EUNITE competition only contain the daily maximum load data and the daily average temperature data, which cannot be used for building (i.e., training) the load forecasting model consisting of significant abnormal peak loads. Recently, a least square method is used to predict the monthly peak load based on the historical

load data [17], which also has difficulty predicting the abnormal peak load with good accuracy. In summary, there is no study for developing probabilistic forecasting models of uncertain building peak loads, and available deterministic models have difficulties in forecasting abnormal peak loads with large fluctuations and little correlation with weather condition.

Due to the stochastic characteristics of building peak load and external impacts such as weather conditions, the building load profile will exhibit volatile characteristics. Hence, a valuable prediction scheme should offer the building load prediction as the form of probability distribution rather than a fixed value. This research aims at developing a probabilistic load forecasting method for individual buildings, whose uncertainty mainly results from uncertain weather conditions and uncertain peak load. Compared with the existing probabilistic building electric load forecasting models, the major novelty and contributions are listed as follows.

- (1) The probabilistic load forecasting model is developed for real-time electric load forecasting of individual buildings considering the real-time weather forecasting uncertainty and uncertain peak load, which is vital for adaptive online decision-making and is essential for the robust control of building energy systems.
- (2) The statistic models for probabilistic abnormal peak load forecasting (i.e., the occurrence and magnitude models of the abnormal peak load) are developed to forecast the uncertain peak load, which are applicable for predicting the abnormal peak loads with significant fluctuations.
- (3) A hybrid method, integrating the linear regression method and the local outlier factor method, is proposed to decompose the load measurement dataset, which shows better decomposition results than using either of them.
- (4) The results from model validation and case studies based on the measured load data show that the developed probabilistic abnormal peak load forecasting and probabilistic normal load forecasting exhibit satisfactory accuracy.

In this study, the artificial neural network (ANN) is adopted to build the probabilistic normal load forecasting model. Eight years of

measured electric load data in an educational building in Hong Kong are used to train and validate the proposed models. Real-time case application studies are conducted to compare different means of using the probabilistic weather forecasts to forecast the probabilistic normal loads. This paper presents the probabilistic load forecasting model, the processing method of the probabilistic weather forecasts, the load dataset decomposition method, the model training, and validation results as well as the real-time case studies.

2. Probabilistic load forecasting model and approach

In this paper, the term ‘load’ represents the building electric load. The term ‘normal load’ represents the load which has no significant deviation from its neighboring loads in magnitude. The term ‘abnormal load’ represents the load which has a significant deviation from its neighboring loads in magnitude. The term ‘abnormal peak load’ represents the load has a significant and positive deviation from its neighboring loads in magnitude. The term ‘peak abnormal differential load’ is defined as the differential between the abnormal peak load and the normal load at a time step.

2.1. The modeling approach

The overall approach of the proposed probabilistic load forecasting model is illustrated in Fig. 1. It comprises two basis components for two major functions: the probabilistic normal load forecasting and the probabilistic peak abnormal differential (PAD) load forecasting. For the probabilistic normal load forecasting, a model is built using the artificial neural network (ANN), which uses the probabilistic weather forecast and the calendar information. For the probabilistic PAD load forecasting, statistical methods are used to quantify two major probabilistic characteristic parameters, i.e., the probability of the occurrence and the probability distribution of magnitude. The probabilistic forecasts of the occurrence and magnitude of the PAD load are generated based on the calendar information.

In this study, the probabilistic load forecast is assumed to contain the probabilistic forecasts of the normal load and the peak abnormal differential load as only the abnormal peak load is concerned in peak

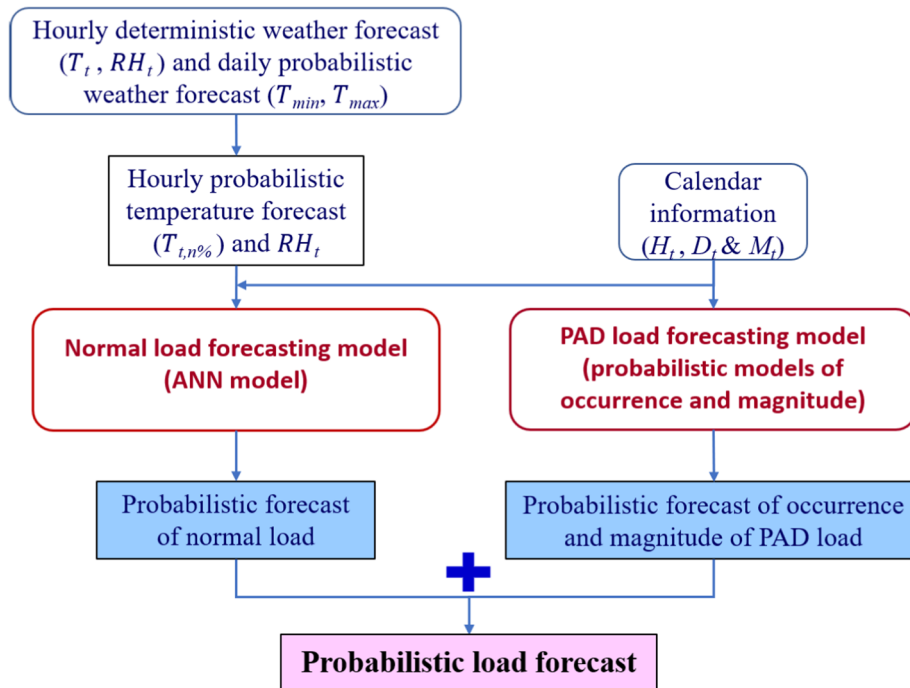


Fig. 1. Components and steps of the proposed probabilistic load forecasting model.

demand management. In an actual load simulation of a particular moment, the probabilistic load forecast might equal to the normal load forecast (absence of abnormal load) or the normal load forecast plus the peak abnormal differential load forecast.

2.2. Probabilistic normal load forecasting model

The electricity consumption in commercial buildings is dominantly affected by outdoor weather conditions and space usage. The relationship between the load and the outdoor weather variables can be described using various models, e.g., the linear regression model. But the influence of space usage on the electric load is difficult to quantify as space usage needs to be described by many variables. However, the calendar information potentially reflects space usage schedule, e.g., the operation schedule of a building and schedule of uses/occupants. Hence, the ANN, a data-driven method, is adopted to model the relationship between the load and the outdoor weather variables together with calendar variables.

2.2.1. Modelling of ANN

The ANN is a self-adaptive machine learning method with few priori assumptions, which is also a universal functional approximator [18]. ANN model extracts the non-linear relationship among the model inputs by different network learning mechanisms using training data [19]. ANN modeling carries out the required output starting the input vectors without considering the assumption of any determinate relationship between the inputs and the output [20]. Hence, ANNs work as “black box” employing distinct features such as input, hidden and out layers of neurons, training functions for the learning process form a set of past data, transfer functions between layers that allow information flow [21]. There are different types of ANNs, among which the back-propagation (BP) neural network is the most popular one [22] and used in this study. BP neural network uses the gradient method to change the weight values and bias values quickly to reduce errors. The typical BP neural network contains three layers, the input layer, the hidden layer, and the output layer. This section describes the process to build the deterministic normal load forecasting model using the three-layer BP neural network. The BP neural network contains many learning algorithms, and it is difficult to select an optimal one for a certain problem. In this study, the Levenberg-Marquardt (LM) learning algorithm is selected due to its higher computational speed in the convergence of the network training compared with other optimization algorithms. Like other optimization algorithms, it is difficult to distinguish whether the LM reaches the global minimum or not. The LM method is selected in this study due to facts that it offers a better learning process and is one of the most widely applied algorithms [23]. In fact, if the test accuracy of the ANN model using LM is acceptable, the difference between the local minimum and global minimum would be minimal and can be ignored.

The selected ANN structure and algorithm are used for the predictive ANN model to forecast the hourly deterministic normal load. This ANN structure contains three layers, the input layer, the hidden layer and the output layer. The detailed ANN model settings are described as follows.

Input layer: In the ANN model, inputs are the outdoor dry-bulb temperature (T_d), relative humidity (RH_d), and the calendar variables (i.e., hour of the day (H_d), day of the week (D_d), and month of the year (M_d)). In this study, only the on-peak time is considered. Therefore each calendar variable has certain ranges (H_d (9, 10, ..., 20), D_d (1, 2, ..., 6) and M_d (1, 2, ..., 12)).

Hidden layer: In the BP network, the number of hidden neurons determines how well a problem can be learned. The trial and error approach is usually used to obtain the right number of hidden neurons since there is no science to this issue. In general, the number of hidden neurons might be evaluated using the following empirical formula [24]:

$$m = \frac{1}{2}(\text{inputs} + \text{outputs}) + \sqrt{\text{number of training patterns}} \quad (1)$$

Output layer: The output of the ANN model is the normal load (L_d) at the same time series as set in the inputs.

Learning algorithm: The Levenberg-Marquardt (LM) learning algorithm is chosen, which is known as *trainlm* in the MATLAB toolbox.

Training and testing samples: The historical datasets of the input and output variables are used as the training and testing samples with 7/8 and 1/8 of the datasets, respectively. In the simulation tests of this study, the proposed ANN model is built in the MATLAB environment using the Neural Network Toolbox.

When the ANN deterministic load forecasting model is available, the probabilistic normal load forecast is generated by importing the probabilistic forecast of the input variables into the model. As the outdoor dry-bulb temperature is one of the most dominant contributors to the building electricity consumption among all the input variables and its probabilistic forecast is available from the weather forecast, the probabilistic forecast of the temperature is chosen to quantify the probability distribution of the normal load forecast in this study.

2.2.2. Generation of hourly probabilistic temperature forecast

The hourly probabilistic forecast of the outdoor dry-bulb temperature is generated based on the daily probabilistic maximum/minimum temperature forecasts and the hourly deterministic temperature forecast. The Hong Kong Observatory (HKO) provides the probabilistic forecasts of the daily minimum and maximum temperatures in the web plot format respectively (see Appendix A), which are digitized into time series data, in this study, using WebPlotDigitizer [25]. The probabilistic forecast of the daily minimum (or maximum) temperature with a particular value (i.e., $T_{min} = T_i$) represents the probability (in percentage) of the daily minimum (or maximum) temperature to occur in the range of plus/minus one degree around that temperature value (i.e. [$T_i - 1$, $T_i + 1$]).

$$F_{T_{min}}(T_i) = P(T_{min} \leq T_i + 1) - P(T_{min} \leq T_i - 1) \quad (2)$$

where $F_{T_{min}}$ is the forecasted probability of the daily minimum temperature with a particular value from the HKO weather forecast. P is the probability of a variable. T_{min} represents the probabilistic forecast of the daily minimum temperature, and T_i is a particular value of T_{min} . Meanwhile, the probabilistic forecast of the daily maximum temperature from the HKO weather forecast is expressed in the same way as Eq. (2).

The aforementioned probabilistic forecast of the daily minimum temperature from the HKO weather forecast is different from the commonly used cumulative probability. Hence, the probabilistic forecast of the daily minimum temperature is firstly transferred into the cumulative distribution function (CDF) of the daily minimum temperature forecast with a one-degree temperature interval. Then both the 1–99% quantile CDFs of the daily probabilistic minimum temperature forecast and the daily probabilistic maximum temperature forecast are generated using the linear interpolation among the previously transferred CDF (with one-degree temperature interval). Afterward, the hourly probabilistic temperature forecast at the $n\%$ quantile at the hour h ($T_{h,n\%}$) is generated using the linear interpolation on the basis of the estimated 1–99% quantile CDFs of the daily minimum and maximum temperature forecasts ($T_{min,n\%}$ and $T_{max,n\%}$), the daily deterministic minimum and maximum temperature forecasts (T^{min} and T^{max}) and the hourly deterministic temperature forecast at the hour h (T_h).

With the estimated hourly probabilistic temperature forecast among all the quantiles (i.e., $n\% \in (1\%, \dots, 99\%)$) at the hour h , the hourly probabilistic normal load forecast is generated at each quantile accordingly. To be specific, for each predicted quantile $T_{h,n\%}$ of the temperature, the ANN load forecasting model is performed by replacing the deterministic value T_h with the estimated probabilistic value $T_{h,n\%}$, and a probabilistic normal load $L_{h,n\%}$ is obtained accordingly.

2.3. Probabilistic peak abnormal differential load forecasting model

As aforementioned, the normal load is highly related to the outdoor weather conditions and the calendar information. However, the PAD load is irrelevant to the outdoor weather conditions but partially related to the calendar information. Besides, the PAD load usually occurs with a small probability, which means that the historical data of the PAD load are quite limited. Hence, a few years' data of the PAD load are required to model the relationship of the PAD load characteristics (occurrence and magnitude) with calendar information. In this section, the probability distribution of the PAD load occurrence is firstly modeled using statistical analysis. Secondly, the probability distribution of the PAD load magnitude is modeled using one of the four commonly-used probability density functions (PDF) as listed later in this section, which is selected based on their fitness.

2.3.1. Probabilistic model of occurrence

The PAD load occurrence is highly related to the hour of the day, and the month of the year. Hence, the probabilistic model of the PAD load occurrence is used to forecast the probability of the PAD load occurrence ($P_{PAD,pre}^{m,h}$) based on the empirical probability of the PAD load occurrence ($P_{PAD}^{m,h}$), as given by Eq. (3).

$$P_{PAD,pre}^{m,h} = P_{PAD}^{m,h}(\text{month: } m = 1 \cdots M, \text{ hour: } h = 1 \cdots H) \quad (3)$$

where $P_{PAD,pre}^{m,h}$ represents the predicted probability of the PAD load occurrence at a specific time in a period. $P_{PAD}^{m,h}$ represents the empirical probability of the PAD load occurrence based on the historical data. The superscript m and h represent the month and the hour, respectively. In this study, the model covers the office hours (12 h a day) of an entire year (i.e., $M = 12$, $H = 12$).

This probabilistic model of occurrence is built based on the assumptions that the PAD load occurs with the same probability at certain office hour of all work days in a month. Here, a “relative PAD load occurrence frequency” is defined as the percentage of PAD load occurrence times at certain office hour in work days among the total times of PAD load occurrence in the period of training (or historical) data. A “monthly PAD load occurrence probability” is defined as the ratio of the total times of PAD load occurrence to the total number of office hours in a month. Therefore, the PAD load occurrence probability at a certain hour h in a month m is calculated by multiplying the relative PAD load occurrence frequency and the monthly PAD load occurrence probability.

2.3.2. Probabilistic model of magnitude

The probability distribution of the PAD load magnitude is firstly fitted using four commonly-used PDFs: i.e., Normal distribution, Gamma distribution, Weibull distribution and Lognormal distribution [26]. To be specific, the training data of the PAD load magnitude (x) are used to fit its probability distribution ($f(x)$) using the four PDFs. The PDF offering the best fitness is selected as the probability distribution forecast model of the PAD load magnitude. Detailed descriptions of the four PDFs are presented in Table B.1 in Appendix B.

2.4. Model parameter identification

The parameters of the deterministic ANN load forecasting model and the probabilistic PAD load magnitude model are identified based on the model identification approaches and the preprocessed training datasets. These model identification approaches are presented in this section, while the preprocessing of the raw training data is presented in the next section.

When training the ANN model, the weights and biases of the network are tuned to optimize the network performance (i.e., minimize the mean square error between the network outputs and the target outputs/training data) using the steepest descent training algorithm [27] by

adopting the Neural Network Toolbox in MATLAB.

The mean square error mse is calculated by Eq. (4). Where, $\mathbf{z}^a$ is the actual target output vector. $\mathbf{z}$ is the network output vector.

$$mse = \frac{1}{N} \sum_{i=1}^N (\mathbf{z}_i^a - \mathbf{z}_i)^2 \quad (4)$$

In the training/fitting of the probabilistic PAD load magnitude model, the parameters defining Normal distribution, Gamma distribution and Lognormal distribution of the PAD load magnitude are estimated based on the commonly-used the maximum likelihood method as presented in Table C.1 in Appendix C. As for Weibull distribution, the parameters might be estimated using several methods such as the moment method, the graphical method and the maximum likelihood method, among which the maximum likelihood method performs the best [28] and is also chosen in this study as listed in Table C1 in Appendix C.

After the parameters of the four fitted PDFs are identified, the Kolmogorov-Smirnov (K-S) test is introduced as the error metric to evaluate the fitness of each PDF [29]. The K-S error is based on the largest vertical difference between the $C(l)$ and $O(l)$, as expressed by Eq. (5).

$$K - \text{Error} = \max |C(l) - O(l)| \quad (5)$$

where C is the calculated cumulative distribution function for the PAD load below a particular value, based on the fitted PDF. O is the cumulative distribution function for the PAD load below the same value, based on the observed PAD load data. l is the concerned value of the PAD load. The smaller the K-S error is, the better the fitness of the PDF is.

3. Decomposition of load dataset and data pre-processing

To identify the model parameters, the historical load dataset (i.e., training dataset) are firstly decomposed into the normal load dataset and the abnormal load dataset (e.g., the PAD load dataset). Then all the datasets of the model variables are normalized into the range [0, 1]. The model forecasts (in the normalized scale) are denormalized into their original scale in the testing section.

3.1. Decomposition of load dataset

In this section, two methods are hybridized and used to decompose the original load dataset into the normal load dataset and the abnormal load dataset. They are the linear regression (LR) method and the local outlier factor (LOF) method. The LR method can decompose the abnormal loads which have little correlation with the outdoor weather conditions among the whole load dataset, but it has weakness in detecting the abnormal loads in the local time series load dataset (such as load data in a day). The LOF method can well detect the discontinuous abnormal load from its neighboring loads, but it might fail when a group of abnormal loads occurs continuously. Hence, taking the advantages of both methods, a hybrid method integrating the LR and the LOF is proposed to decompose the original load dataset.

3.1.1. Decomposition for abnormal and normal load datasets

Assuming that Set $\mathbf{A}$ is the abnormal load dataset decomposed using the LR method and Set $\mathbf{B}$ is the abnormal load dataset decomposed using the LOF method, the combined abnormal load dataset is the union of the two sets, as expressed by Eq. (6). After the combined abnormal load dataset is decomposed from the original load dataset, the gaps in the original time series load dataset are filled by the “recovered normal loads” using the linear interpolation method, as expressed by Eq. (7), for the convenience of using the dataset.

$$\mathbf{A} \cup \mathbf{B} = \{x: x \in \mathbf{A} \text{ or } x \in \mathbf{B}\} \quad (6)$$

$$\frac{L_{gap,t} - L_{t-i}}{i} = \frac{L_{t+j} - L_{gap,t}}{j} \quad (7)$$

where $L_{gap,t}$ is the gap in the original time series load dataset at the time t after load dataset decomposition. L_{t-i} and L_{t+j} are the two nearest neighboring time series loads from the $L_{gap,t}$. The original time series load dataset with the abnormal loads replaced by their recovered normal loads is regarded and used as the updated normal load dataset.

After the decomposition of the load dataset, the abnormal differential load is defined as the deviation between the abnormal load and the updated normal load at the same time. Then, the dataset of the abnormal differential load with a positive value is regarded as the peak abnormal differential load dataset.

3.1.2. Decomposition of load dataset using linear regression

In the decomposition of the load dataset using the LR method, the sensitivity analysis is firstly conducted using the Pearson correlation [30] between the load and each weather variable (dry-bulb temperature, relative humidity, and enthalpy). Next, the datasets of the load together with the selected sensitive weather variable are clustered according to the hour of the day, and then the linear regression model is built between the two variables in each cluster. Clustering the load data by time period is to consider the fact that the dominating factors on the load are not only weather but also the pattern of activities which are regarded to be time-dependent. Finally, the Kernel density estimation (KDE) function is used to analyze the residuals between the actual loads and the estimated loads in each cluster and decompose the abnormal loads.

The enthalpy represents the combined weather effects of the outdoor dry-bulb temperature and humidity. It is worth noticing that the sensitivity analysis was conducted between the load and each weather variable (i.e., T , RH , and h) in k th order based on the Pearson correlation coefficient, which is widely used in the measure of the linear relationship between two variables [31]. The enthalpy is found to be the best representative weather variable, as shown in Table 1, when only one is to be selected. Table 1 shows that the enthalpy in the second order is the most sensitive to the load, with the largest Pearson coefficient of 0.914. Hence, the linear regression model is built between the load and the enthalpy in the second order in each cluster.

To obtain the estimates of the loads which are highly correlated to the outdoor weather conditions, a linear regression model in each cluster is established as expressed by Eq. (8). Then, the residual ε_t between the actual load and the estimated load is calculated using Eq. (9).

$$L_t' = a_0 + a_1 h_t^2 + a_2 h_t \quad (8)$$

$$\varepsilon_t = L_t - L_t' \quad (9)$$

where L_t' represents the estimated hourly load (kVA) at the time t using the linear regression model. a_0 , a_1 , and a_2 are the constant coefficients, which can be identified by the least square method using the historical data. L_t is the hourly actual load (kVA) at the time t .

To decompose the abnormal loads from normal loads, the actual loads with large discrepancy with the estimated loads are separated by analyzing the residuals using the Kernel density estimation function. The Kernel density estimation function is a non-parametric estimate of the distribution shape [32], as expressed by Eq. (10). Where, the bandwidth h_{bw} is optimized and identified using Eq. (11) [33].

$$\hat{f}_{h_{bw}}(\varepsilon) = \frac{1}{nh_{bw}} \sum_{i=1}^n K\left(\frac{\varepsilon - \varepsilon_i}{h_{bw}}\right) \quad (10)$$

$$h_{bw} = \left(\frac{4\hat{\sigma}^5}{3n}\right)^{\frac{1}{5}} \quad (11)$$

where ε_1 , ε_2 , ..., and ε_n are the random residual samples. n is the sample size. $K(\cdot)$ is the Kernel smoothing function. h_{bw} is the bandwidth. $\hat{\sigma}$ is the standard deviation of the residual samples. The actual loads, with their corresponding residuals out of the 95% confidence interval of the KDE function, are decomposed as the abnormal loads in each cluster. Finally, the decomposed abnormal loads of all clusters are united as the total abnormal load dataset.

3.1.3. Decomposition of load dataset using local outlier factor

In the decomposition of load dataset using the local outlier factor (LOF) method, the LOF value of each time series load is firstly calculated. Then all LOF values of loads are analyzed using the empirical cumulative distribution function (ECDF) to decompose the abnormal loads.

The LOF method is a classic outlier detection strategy [34]. With the LOF method, the anomaly degree of the load L_t in the dataset is measured with the LOF value [35], as given by Eq. (12).

$$LOF(L_t) = \frac{\sum_{i \in N_k(L_t)} lrd_k(i) / lrd_k(L_t)}{|N_k(L_t)|} \quad (12)$$

where $lrd_k(i)$ is the local density the object i from its neighbors. $lrd_k(L_t)$ is the local density of the load L_t in its k -distance neighborhood $N_k(p)$. $|N_k(p)|$ is the size of $N_k(p)$. k is the number of the minimum points in the neighborhood, typically set to be not less than 10 [34]. $LOF(L_t)$ reflects the degree of the load L_t as a local outlier. A value of approximately 1 indicates that the load at the time t is very close to its neighbors and thus not an abnormal load, while values significantly larger than 1 indicate abnormal loads.

To decompose the abnormal loads with the larger LOF values, all loads' LOF values are analyzed with the empirical cumulative distribution function [36]. Let $(x_1, \dots, x_n)$ be the random values (i.e., loads in this study) with the common cumulative distribution function $F(l)$. Then the empirical cumulative distribution function is defined by Eq. (13).

$$\hat{F}_n(l) = \frac{\text{number of elements in the sample} \leq l}{n} \quad (13)$$

where n is the number of all the random values (loads). l is a particular value of the random values (loads). When n approaches ∞ , $\hat{F}_n(l)$ approaches $F(l)$. The LOF value significantly larger than 1 corresponds to the larger cumulative probability, which means that the abnormal loads with the larger LOF values have larger cumulative probabilities. Hence, the actual loads with their LOF values' cumulative probability over 95% are decomposed as the abnormal load dataset.

3.2. Data preprocessing

After the decomposition of the original load dataset, the datasets for model training are normalized using Eq. (14). Where, y is a variable. y_i is a random value of the variable y . $y_{noz,i}$ is the normalized value. $\max(y)$ and $\min(y)$ are the maximum and minimum values in the dataset respectively.

$$y_{noz,i} = \frac{y_i - \min(y)}{\max(y) - \min(y)} \quad (14)$$

Table 1
Computed Pearson correlation coefficients between load and each weather variable.

	T	T^2	T^3	RH	RH^2	RH^3	h	h^2	h^3
Pearson coefficient	0.876	0.880	0.870	0.246	0.211	0.177	0.910	0.914	0.902

4. Results of model training and validation

4.1. Data description

An educational building named Phase 7, of The Hong Kong Polytechnic University in Hong Kong (a sub-tropic climate zone), is selected in the study. It is equipped with different research facilities, laboratories, classrooms, conference rooms, lecture theaters, and staff offices. In Hong Kong, the peak load (demand) of commercial buildings usually occurs during the on-peak time (i.e., from 9:00 a.m. to 21:00p.m. and excluding Sundays together with public holidays), which results in a critical issue for peak demand liming and contributes to the high monthly electricity cost. For simplicity and forecast precision, the historical datasets only during the on-peak time (i.e., office hour) are used for model training and validation tests, and the load spikes from chillers' starting in the morning before 8 a.m. are not considered in this study. This choice is also largely due to the fact that these load spikes are not the effective causes of actual monthly peak loads based on analyzing the practical load profiles of many buildings. In the training and validation tests, the historical data of the load, weather variables, and calendar variables were collected at the hourly time interval from Jan 2009 to Dec 2016. The load profile of Phase 7 shows a seasonal variation and uncertain peak loads in each month, as illustrated in Fig. 2.

4.2. Model training and results

4.2.1. Decomposition of load dataset

In the decomposition of load dataset using the LR method, the historical data of load and enthalpy from Jan 2009 to Dec 2016 were classified into 12 clusters (based on office hour of working days) and then the coefficients of the LR model in each cluster were identified, as listed in Table 2.

The residuals between the actual loads and the estimated loads in each cluster were analyzed using the Kernel density estimation function. The optimized bandwidth of the KDE function in each cluster was set as 25 according to the optimization formula. All the actual loads with their corresponding residuals out of the 95% confidence interval of the KDE function were selected as the abnormal loads (Set A). Fig. 3 shows the histograms and KDE functions of residuals between the actual loads and the estimated loads at selected hours (i.e., 9:00 and 14:00). It can be seen that the estimated KDE functions well matched the histograms of the load residuals in both of the two clusters.

The historical load data from Jan 2009 to Dec 2016 were also used in the decomposition of load dataset using the LOF method. The LOF value of each hourly load was calculated with the parameter k (set as the recommended value 10). Then all LOF values were analyzed with the empirical cumulative distribution function, and the actual loads with their LOF value's empirical cumulative probability over 95% were regarded as the abnormal loads (Set B). Fig. 4 shows the histograms and empirical cumulative distribution functions of the LOF values of all loads in selected years (i.e., 2009 and 2013). It can be seen that most of the LOF values of loads were close to 1.

After load dataset decomposition using each of the LR, LOF and hybrid methods, the ratios of the number of the decomposed abnormal loads to the number of all the loads using these methods are listed in Table 3. It can be seen that the ratio from using the hybrid method integrating LR and LOF was 0.068, much larger than that from using either method, which indicates the hybrid method is more sensitive and robust.

The results of the load datasets decomposition during the office hour in 2013 are shown in Fig. 5 for illustration. It can be seen that most extremely large and small loads were decomposed as abnormal loads by both the LR and the LOF methods.

4.2.2. Training of ANN model

To train the ANN model, the historical datasets of the normal load, weather variables, and calendar variables from Jan 2009 to Dec 2015 were used. The number of hidden neurons in the hidden layer was set as 12 according to Eq. (1) and the trial and error. Besides, the mean square error was set as 0.001, the learning rate was set as 0.5, and the iteration times were set as 1000. The identified weight values and the bias values of the ANN model are listed in Table D.1 in Appendix D.

4.2.3. Training of probabilistic magnitude model

In the training of the probabilistic model of the peak abnormal differential (PAD) load magnitude, the PAD load data from Jan 2009 to Dec 2015 were used to train/fit the probability distribution of the magnitude using the four alternative PDFs. The identified parameter values of the fitted PDFs of the PAD load magnitude are listed in Table E.1 in Appendix E.

Table 4 shows the statistical errors of the trained PDFs of the PAD load magnitude data. The Lognormal distribution fits the observed data best among the four PDFs, with the smallest K-S error of 0.0398. Therefore, the Lognormal distribution ($\varphi = 4.5$, $\gamma = 0.7$) is selected for the forecasted probability distribution of the magnitude in the later years.

4.3. Model validation and results

4.3.1. Performance evaluation criteria

To evaluate the deterministic normal load forecasting model, the commonly-used mean absolute percentage error (MAPE) indicator, mean absolute error (MAE) indicator and root mean squared error (RMSE) are adopted, as given by Eqs. (15)–(17). Where, L_i is the measured load (kVA). $\hat{L}_i$ is the forecasted load (kVA). i is a certain number. n is the length of the test time series.

$$\text{MAPE} = \frac{1}{n} \sum_{i=1}^n \left| \frac{L_i - \hat{L}_i}{L_i} \right| \quad (15)$$

$$\text{MAE} = \frac{1}{n} \sum_{i=1}^n |L_i - \hat{L}_i| \quad (16)$$

$$\text{RMSE} = \sqrt{\frac{1}{n} \sum_{i=1}^n (L_i - \hat{L}_i)^2} \quad (17)$$

To evaluate the PAD load magnitude model, the deviations between the observed data and their forecasts are assessed using the K-S error as shown earlier in Eq. (5).

4.3.2. Validation of ANN deterministic load forecasting model

In the validation test of the ANN deterministic load forecasting model, the historical datasets of the normal load, weather variables, and calendar variables in 2016 were used. The calculated monthly MAPes, MAEs and RMSEs of the test results are listed in Table 5. The

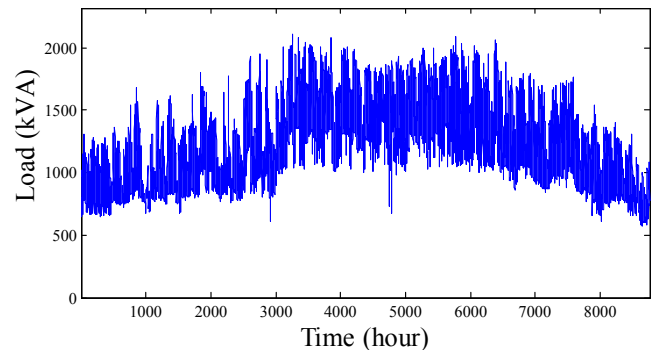


Fig. 2. The electric load profile of Phase 7 in 2013.

Table 2
Identified coefficients of the linear regression model in each cluster.

Coefficient	Cl ₁	Cl ₂	Cl ₃	Cl ₄	Cl ₅	Cl ₆	Cl ₇	Cl ₈	Cl ₉	Cl ₁₀	Cl ₁₁	Cl ₁₂
a_0	874	842	868	846	846	850	804	803	821	828	798	778
a_1	0.12	0.09	0.09	0.09	0.09	0.08	0.07	0.07	0.09	0.12	0.12	0.11
a_2	1.91	5.33	5.14	5.21	4.90	6.00	8.08	7.89	5.96	3.14	2.89	2.93

Remark: Cl₁ represents the cluster of datasets at 9:00 and Cl₁₂ represents the cluster at 20:00.

average MAPE was 5.0%, the average MAE was 77.8 kVA, and the average RMSE was 108 kVA in the year. The minimum MAPE was 2.9%, which occurred in July, the minimum MAE was 46 kVA, which occurred in January, and the minimum RMSE was 54 kVA, which occurred in January. This shows the forecast accuracy of the deterministic load forecasting model is satisfactory.

The detailed load forecast in July and December 2016 and the comparison with actual measurements are shown in Fig. 6, which are the best and the worst results respectively. It can be seen that load forecasts in both months matched the actual loads with satisfactory agreement, while the agreement in July was better than that in December. It is because that in Hong Kong December is in the transition season and the weather condition fluctuates significantly.

4.3.3. Validation of probabilistic models of occurrence and magnitude

To validate the probabilistic model of the peak abnormal differential (PAD) load occurrence, the PAD load data in 2016 were used. The forecasts of the relative PAD load frequencies (i.e., percentages) at different office hours of work days in 2016 are shown and compared with the actual data in Fig. 7. It can be seen that the forecasts of the relative frequencies matched the actual data very well during most hours while the forecasts at 9:00, 19:00 and 20:00 have relatively larger deviations. With more PAD load data, the forecast accuracy of the relative frequency can be improved.

To validate the probabilistic model of the PAD load magnitude, the PAD load data in 2016 were also used. The K-S error between the forecasted distribution and the observed distribution of the PAD load in 2016 was 0.09, showing a very satisfactory accuracy. The forecasted and observed probability densities of the PAD load magnitude are shown in Fig. 8. It can be seen that most of the forecasted probability densities of the PAD load magnitude matched well the observed ones.

5. Real-time application case studies

In the real-time application case studies, the 9-day interval probabilistic normal load forecasts were generated using 9-day interval weather forecast which is provided and updated daily by the Hong

Kong Observatory (HKO).

To evaluate the probabilistic forecast models, the sharpness, resolution and reliability of forecasts are usually considered [4], which are also considered in this study to evaluate the probabilistic forecast model of the normal load. The pinball loss function [37] is used to calculate the quantile scores of the probabilistic load forecast results. The pinball loss for the load at each quantile and each hour is calculated using Eq. (18).

$$\text{Pinball}(L_{t,q\%}, L_t, q\%) = \begin{cases} q\%(L_t - L_{t,q\%}) & L_t \geq L_{t,q\%} \\ (1 - q\%)(L_{t,q\%} - L_t) & L_t < L_{t,q\%} \end{cases} \quad (18)$$

where $q\%$ is the target quantile of the quantile load forecast. The mean of the pinball losses across all the quantiles and all the forecasted hours in a month is regarded as the pinball score for the probabilistic load forecast. A lower pinball score indicates a better probabilistic forecast.

5.1. Case 1 – October (using weather forecasts on four days)

The 9-day weather forecast data in October 2017 reported by the HKO in four days (i.e., 30/Sept, 9/Oct, 18/Oct and 27/Oct) were used to compose the probabilistic weather forecasts of all days in the month. The weather forecast reported by the HKO includes the probabilistic forecasts of the daily maximum/minimum temperatures and the deterministic forecasts of the hourly temperature and relative humidity. The hourly probabilistic temperature forecasts, as shown in Fig. F.1 in Appendix F, were obtained by processing these available temperature forecast data using the proposed method in Section 2.2.2.

These hourly probabilistic temperature forecasts and the hourly deterministic relative humidity forecasts during office hours in October were used by the ANN model as the inputs to generate the probabilistic normal load forecasts and the results are shown in Fig. 9. Fig. 9 shows the actual loads, the mean probabilistic normal load forecasts and the 90% prediction intervals in October and the probability density of normal load at an hour. It can be seen that most of the actual loads fell in the 90% prediction intervals. The small figure in the right illustrates

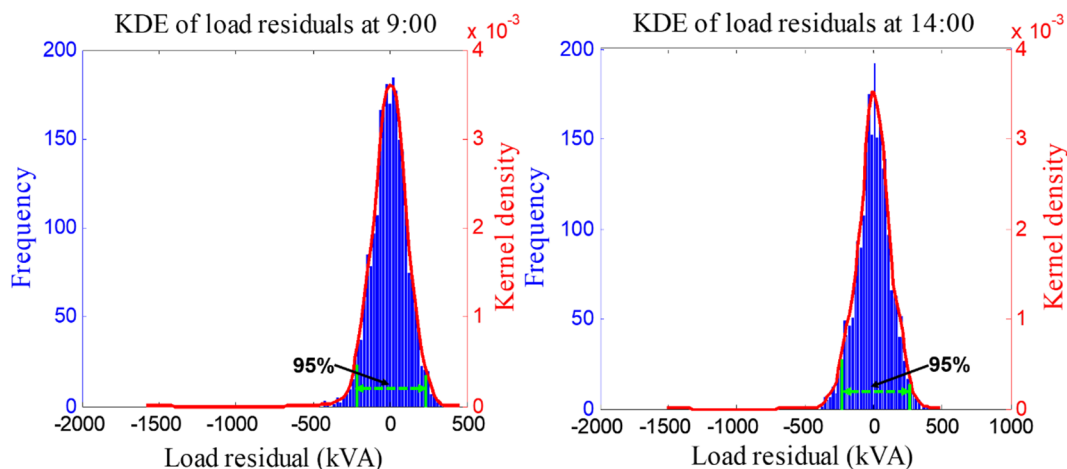


Fig. 3. Histograms and Kernel density estimation functions of load residuals at 9:00 and 14:00.

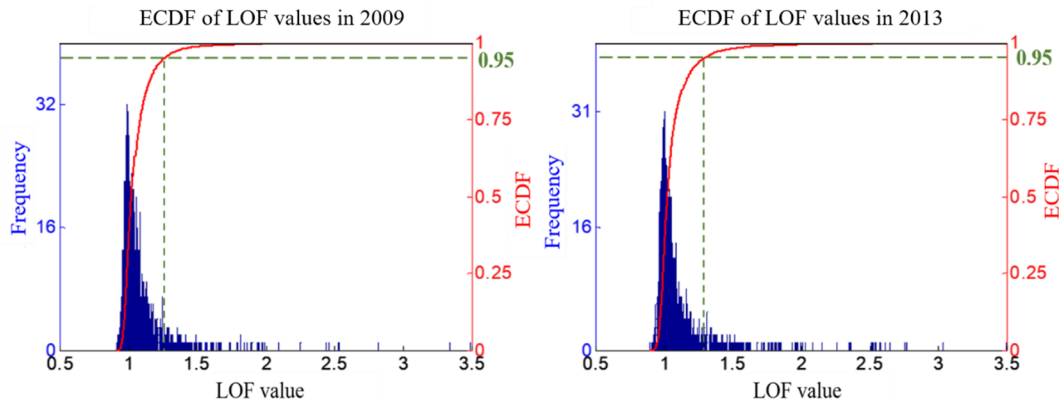


Fig. 4. Histograms and empirical cumulative distribution functions of the LOF values of the loads in 2009 and 2013.

Table 3

The ratios of the number of the decomposed abnormal loads to the number of all the loads using the three methods.

	LR	LOF	Hybrid
ratio	0.036	0.035	0.068

the probability distribution of the forecasted normal load at 14:00 on 16/Oct. The MAPE, MAE and RMSE between the mean probabilistic normal load forecasts and the actual loads in October were 0.084, 147 kVA and 170 kVA respectively (listed in Table 6), showing an acceptable forecast accuracy.

5.2. Case 2 – December (using one-day-ahead weather forecasts)

The 9-day weather forecast data reported by the HKO in December 2017 were recorded daily and used to compose the probabilistic weather forecasts of all days in the month. Different from Case 1, the one-day-ahead weather forecasts updated daily were used for load forecasts of every day in the month. It means that the normal load forecasts of a day are obtained using the weather forecasts of that day reported in the previous day. The hourly probabilistic temperature forecasts using the one-day-ahead weather forecasts in December were generated in the same way as that in Case 1, as shown in Fig. F.2 in Appendix F.

Table 4

Statistical errors for the proposed probability density functions.

	Normal	Gamma	Weibull	Lognormal
K-S error	0.1379	0.0573	0.0680	0.0398

These hourly probabilistic temperature forecasts and the hourly deterministic relative humidity forecasts during office hours in December were used by the ANN model to generate the probabilistic normal load forecasts, and the results are shown in Fig. 10. Fig. 10 shows the actual loads, the mean probabilistic normal load forecasts and the 90% prediction intervals using one-day-ahead weather forecasts in December and the probability density of normal load at an hour. It can be seen that most of the mean probabilistic normal load forecasts were close to the actual loads, which was better than the forecast results in Case 1. The small figure in the right illustrates the probability distribution of the forecasted normal load at 14:00 on 15/Dec. In this case, the MAPE, MAE and RMSE between the mean probabilistic normal load forecasts and the actual loads in December were 0.047, 64 kVA and 80 kVA respectively (listed in Table 6), showing a satisfactory forecast accuracy.

5.3. Case 3 – December (using nine-day-ahead weather forecasts)

Different from Case 2, the nine-day-ahead weather forecasts

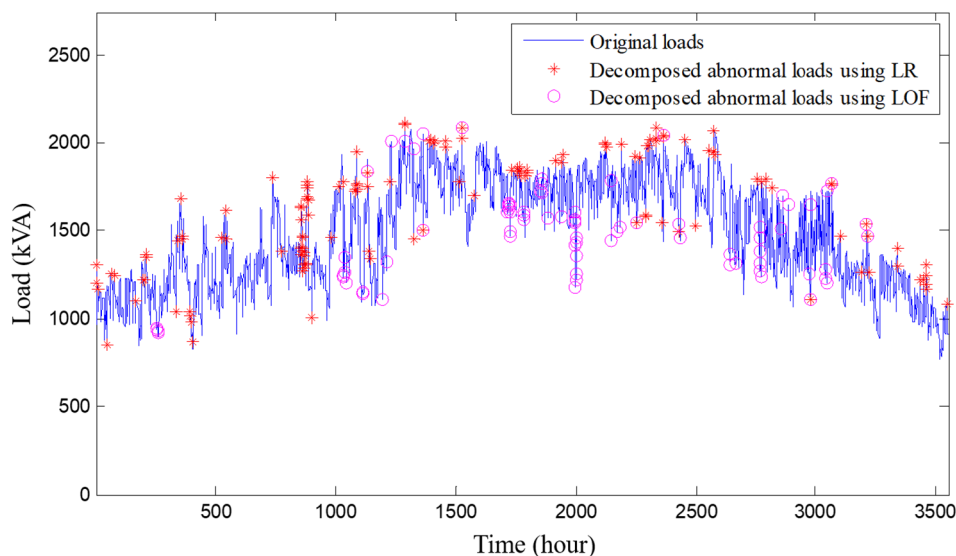


Fig. 5. Load dataset decomposition results during office hours in 2013.

Table 5
MAPEs, MAEs and RMSEs of test results of the ANN model in 2016.

	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sept	Oct	Nov	Dec
MAPE (%)	4.0	4.8	5.4	5.8	6.2	7.0	2.9	3.1	5.0	3.9	6.5	7.0
MAE (kVA)	46	55	70	94	114	142	55	58	80	61	77	82
RMSE (kVA)	54	90	98	144	148	171	79	87	116	101	109	106

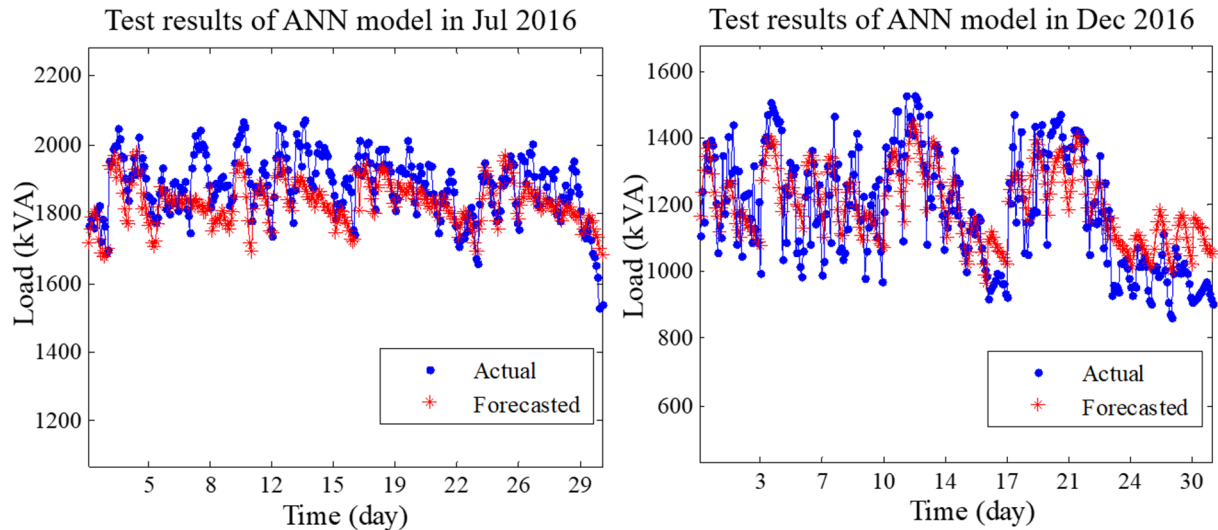


Fig. 6. Load forecast using the deterministic load forecasting model compared with actual measurements during office hours (July and Dec 2016).

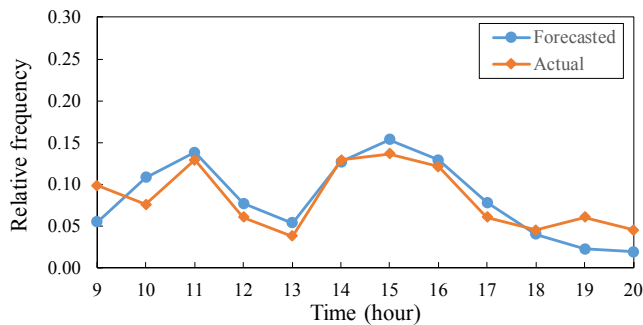


Fig. 7. Forecasted and actual relative frequencies of the peak abnormal differential load at different office hours (2016).

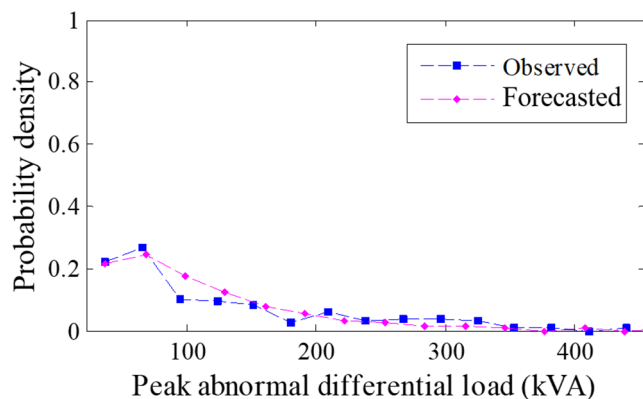


Fig. 8. Comparison between forecasted and observed probability densities of the peak abnormal differential load magnitude (2016).

updated daily were used for normal load forecasts of each day in the month. It means that the normal load forecasts of each day are obtained using the weather forecasts of that day made nine days ago. The hourly

probabilistic temperature forecasts using the nine-day-ahead weather forecasts in December were generated in the same way as that in Case 1, as shown in Fig. F.3 in Appendix F.

These hourly probabilistic temperature forecasts and the hourly deterministic relative humidity forecasts during office hours in December were used by the ANN model to generate the probabilistic normal load forecasts, and the results are shown in Fig. 11. Fig. 11 shows the actual loads, the mean probabilistic normal load forecasts and the 90% prediction intervals using nine-day-ahead weather forecasts in December and the probability density of normal load at an hour. It can be seen that most of the mean probabilistic normal load forecasts were close to the actual loads, which was better than the forecast results in Case 1 but not as good as those in Case 2. The small figure in the right illustrates the probability distribution of the forecasted normal load at 14:00 on 15/Dec. In this case, the MAPE, MAE and RMSE between the mean probabilistic normal load forecasts and the actual loads in December were 0.067, 65 kVA and 82 kVA respectively (listed in Table 6), showing an acceptable forecast accuracy.

5.4. Comparison of forecast accuracies in three cases

To evaluate the probabilistic normal load forecasts in the above three cases, the mean probabilistic normal load forecasts are compared with the actual loads using the MAE, MAPE and RMSE. Besides, the pinball scores of the three cases are calculated and compared. Table 6 presents the detailed performance evaluation metrics for the probabilistic normal load forecasts of the three cases. The smaller the performance evaluation metric is, the more accurate the probabilistic normal load forecasts are. The probabilistic load forecasts in Case 2 shows the best performance with the smallest value in each individual performance evaluation metric.

Fig. 12 shows the percentage of the actual loads falling within the predicted quantile of the probabilistic normal load forecasts in a month period in three cases. It can be seen that the overall predicted quantile bandwidth (1%-99%) in Case 2 was the smallest, which indicates the performance of the probabilistic normal load forecasts in Case 2 is the

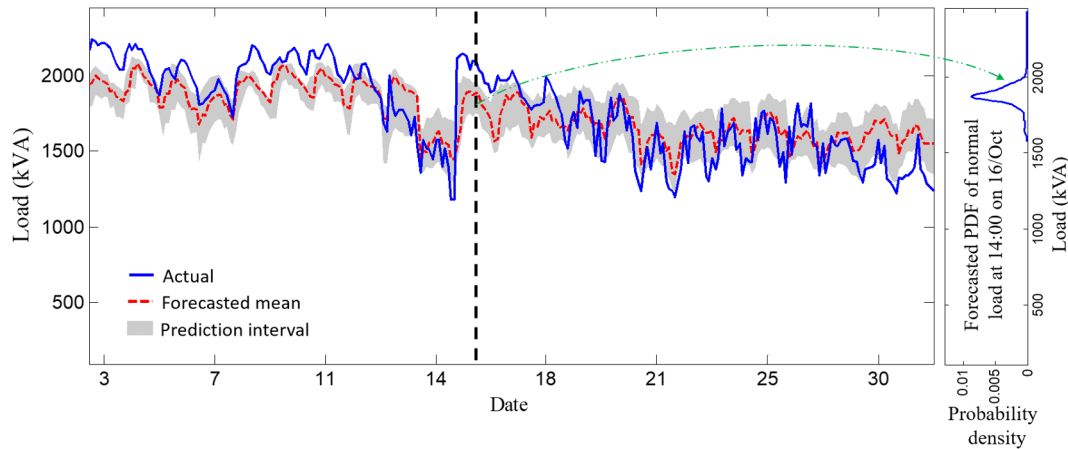


Fig. 9. Probabilistic normal load forecasts of October and the probability density of normal load at one hour. The shading represents the boundaries of the 90% prediction interval.

Table 6

Performance evaluation metrics for the probabilistic normal load forecasts in the three cases.

	Case 1	Case 2	Case 3
MAE_{prob} (kVA)	143	64	65
$MAPE_{prob}$ (%)	8.4	4.7	6.7
$RMSE_{prob}$ (kVA)	170	80	82
Pinball score	58.9	27.6	27.8

Remark: MAE_{prob} is the mean absolute error between the mean probabilistic normal load forecasts and the actual loads, $MAPE_{prob}$ is the mean absolute percentage error between the mean probabilistic normal load forecasts and the actual loads, and $RMSE_{prob}$ is the root mean square error between the mean probabilistic normal load forecasts and the actual loads.

most accurate.

Comparing Case 1 with Case 2, the five probabilistic normal load forecasting accuracy indicators (i.e., MAE_{prob} , $MAPE_{prob}$, $RMSE_{prob}$, the pinball score and the overall predicted quantile bandwidth (1–99%)) of Case 2 were smaller than that of Case 1. It indicates that the probabilistic load forecasts of Case 2 are superior to that of Case 1, and the probabilistic load model can offer better forecasts using the most updated (i.e., one-day-ahead) weather forecasts compared with the case using the nine-day weather forecasts without daily updating.

Comparing Case 2 with Case 3, the five probabilistic normal load forecasting accuracy indicators of Case 2 were smaller than that of Case 3. It indicates that the probabilistic load forecasts of Case 2 are superior

to that of Case 3. It means that the shorter the period of the weather forecast is, the better accuracy the probabilistic normal load forecasts can achieve.

Comparing Case 1 with Case 3, the first three probabilistic normal load forecasting accuracy indicators (i.e., MAE_{prob} , $MAPE_{prob}$, $RMSE_{prob}$) of Case 1 were larger than that of Case 3, due to the fact that the (deterministic) weather forecasts in December were more accurate than that in October. But the last indicator (i.e., the overall predicted quantile bandwidth (1–99%)) of Case 1 was smaller than that of Case 3, indicating that bandwidth of load forecasts was smaller in Case 1.

In general, the probabilistic normal load forecasting accuracy is affected by the probabilistic weather forecasting accuracy from the HKO. Comparison studies for the proposed probabilistic load forecasting approach are expectable and could be conducted in the near future when studies using standard datasets of compatible format (i.e., the probabilistic weather forecast is included) are available.

6. Conclusions

This paper presents a probabilistic load forecasting model, which consists of two probabilistic forecasting models for the normal load and the peak abnormal differential (PAD) load respectively. An ANN model is built for the normal load forecasting, and statistical models are built for the probabilistic occurrence and magnitude forecasting of the peak abnormal differential load. Based on the results of validation tests and case studies, the following conclusions can be made:

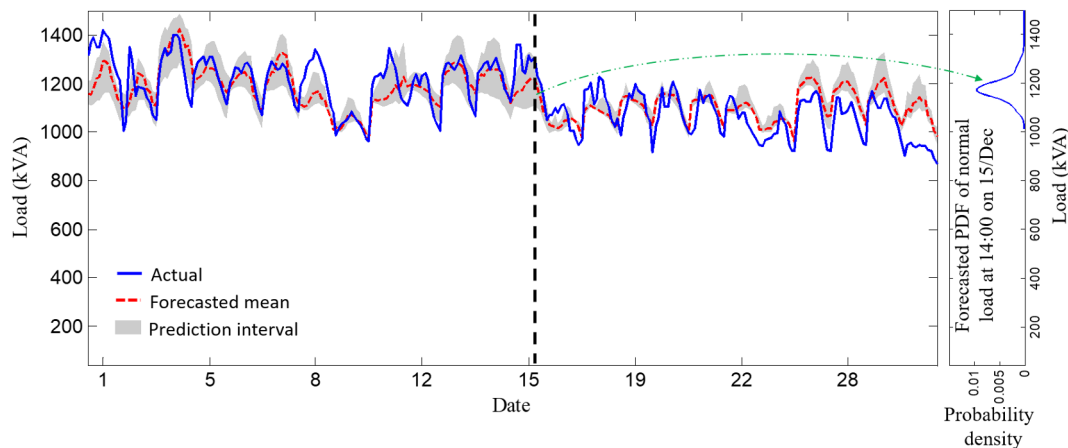


Fig. 10. Probabilistic normal load forecasts of December using one-day-ahead weather forecasts and the probability density of normal load at one hour. The shading represents the boundaries of the 90% prediction interval.

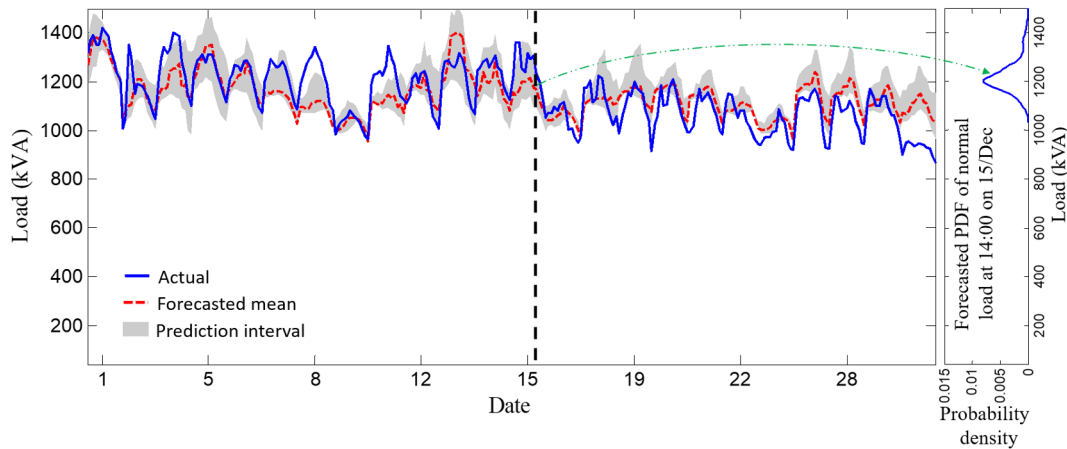


Fig. 11. Probabilistic normal load forecasts of December using nine-day-ahead weather forecasts and the probability density of normal load at one hour. The shading represents the boundaries of the 90% prediction interval.

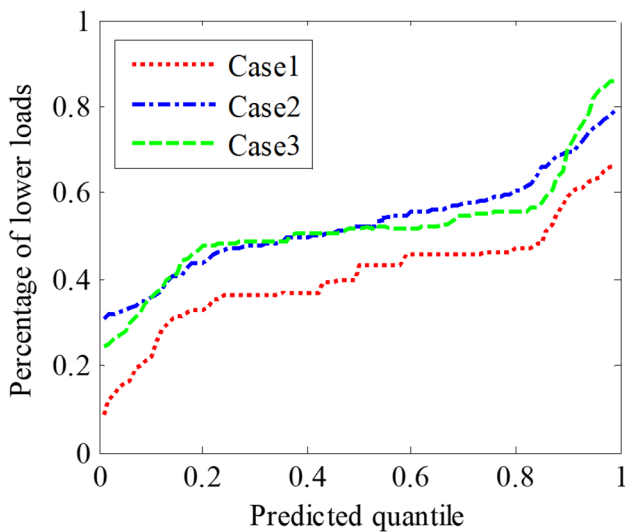


Fig. 12. Comparison of the percentages of observed actual load values falling in the predicted quantile of the probabilistic normal load forecasts in three cases.

- The ANN model for deterministic normal load forecasting is capable of obtaining results with satisfactory accuracy with the average MAPE of 5.0%, the average MAE of 77.8 kVA and the average RMSE of 108 kVA.
- The probabilistic occurrence model of PAD load shows a good performance in forecasting the relative PAD load occurrence frequency.
- The probabilistic magnitude model of the PAD load is fitted based on four commonly-used probability density distributions, among which the Lognormal distribution shows the best fitness of the observed PAD load data. The probabilistic model of the magnitude can well forecast the magnitudes with the Kolmogorov-Smirnov error of 0.09.
- All the three cases achieve acceptable accuracy, and the

probabilistic normal load forecasts based on the most updated (i.e., one-day-ahead) weather forecasts are the most accurate and certain (minimum bandwidth).

In summary, this study develops an approach and schemes for the development of probabilistic load forecasting models by quantifying the uncertainties of the normal load and the abnormal peak load. The proposed probabilistic load forecasting model can successfully forecast the hourly probabilistic load, including the probabilistic normal load and the probabilistic occurrence and magnitude of the abnormal peak load, with satisfactory accuracy.

The developed probabilistic load forecasting model is capable of quantifying the probability of building electric load larger than a certain value at a certain hour, or even on one day or one week. As for the real applications, there would be three main potential uses. The first one is for the optimal limiting control of building peak demand. The probabilistic load forecasts can be applied to adaptively determine the optimal building load set-point/threshold online to minimize the building electricity cost (particularly for the peak demand cost such as under the Hong Kong electricity tariff). The second one is for the day-ahead forecasting of electricity price for the smart grid. The peak load forecast is essential for determining the day-ahead dynamic pricing or critical peak pricing because the peak load is a major concern of a power grid. The third one is for the optimal operation of renewable energy resources. The proposed probabilistic forecasting method can contribute to the optimization of the decisions in the operation of renewable resources under load uncertainties (e.g., the charge or discharge control of electric vehicles).

Acknowledgment

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Appendix A. A web image of the probabilistic forecast of daily minimum temperature

Fig. A.1 shows the captured image of a daily probabilistic minimum temperature forecast from the local weather station in Hong Kong, a subtropical area [38]. It can be seen that the probabilistic forecast of the daily minimum temperature was forecasted in a medium term of 14 days. On each day, the probabilities of the daily minimum temperature in different ranges were estimated and illustrated in different colors.

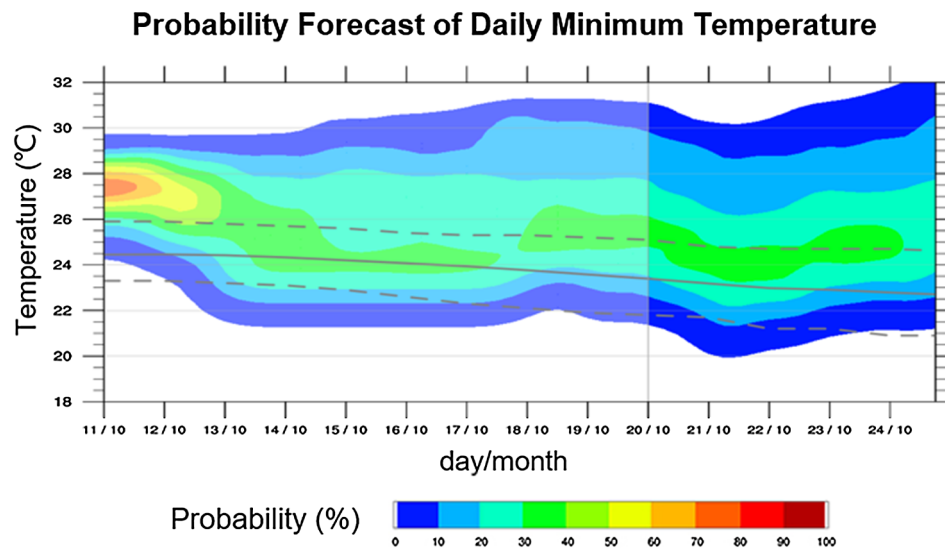


Fig. A.1. An example (web image) of the probabilistic forecast of the daily minimum temperature – HKO weather forecast.

Appendix B. Descriptions of the four commonly-used probability distribution functions

The formula and detailed coefficient descriptions of each of the four commonly-used probability distribution functions are shown in Table B.1.

Table B.1

Detailed descriptions of the four commonly-used probability distribution functions.

Function	Formula	Coefficient description
Normal distribution	$f(x; \mu, \sigma^2) = \frac{1}{\sqrt{2\pi\sigma^2}} e^{-\frac{(x-\mu)^2}{2\sigma^2}}$	μ is the mean parameter, and σ is the standard deviation parameter.
Gamma distribution	$f(x; \alpha, \beta) = \frac{x^{\alpha-1} e^{-\frac{x}{\beta}}}{\beta^{\alpha} \Gamma(\alpha)}$	α is a shape parameter, β is an inverse scale parameter, and $\Gamma(\alpha)$ is a complete Gamma function [39].
Weibull distribution	$f(x; k, \lambda) = \frac{k}{\lambda} \left(\frac{x}{\lambda}\right)^{k-1} e^{-(x/\lambda)^k}$	λ is the scale parameter, and k is the shape parameter [28].
Lognormal distribution	$f(x; \varnothing, \gamma) = \frac{1}{x\sqrt{2\pi\varnothing^2}} e^{-\frac{(\ln(x)-\gamma)^2}{2\varnothing^2}}$	$\varnothing$ is the mean parameter of $\ln(x)$, and γ is the standard deviation parameter of $\ln(x)$ [40].

Appendix C. Parameter identification of the four commonly-used probability distribution functions

The detailed parameter identification formula using the maximum likelihood method detailed descriptions of each probability distribution function are shown in Table C.1.

Table C.1

Detailed descriptions of the maximum likelihood method for parameter identification of the four commonly-used probability distribution functions.

Function	Identified parameter	Description
Normal distribution	$\mu = \frac{1}{n} \sum_{i=1}^n x_i$ $\sigma^2 = \frac{1}{n} \sum_{i=1}^n (x_i - \mu)^2$	Normal distribution parameters are evaluated using the maximum likelihood method.
Gamma distribution	$LL = \ln \prod_{i=1}^n \{h(x_i; \alpha, \beta)\}$ $= \sum_{i=1}^n \ln \{h(x_i; \alpha, \beta)\}$	Gamma distribution parameters are evaluated using the maximum likelihood method which maximizes the logarithm of the likelihood function [39].
Weibull distribution	$k = \left[\frac{\sum_{i=1}^n x_i^k \ln(x_i)}{\sum_{i=1}^n x_i^k} - \frac{\sum_{i=1}^n \ln(x_i)}{n} \right]^{-1}$ $\lambda = \left[\frac{1}{n} \sum_{i=1}^n x_i^k \right]^{1/k}$	Weibull shape and scale parameters are calculated using the maximum likelihood method [28].
Lognormal distribution	$\gamma = \frac{1}{N} \sum_{i=1}^N \ln(x_i)$ $\varnothing^2 = \frac{1}{N} \sum_{i=1}^N [\ln(x_i) - \gamma]^2$	Lognormal parameters are estimated using the maximum likelihood method without an iterative procedure [40].

Remark: The shape and scale parameters are estimated based on the given data, similar to the estimation of the mean parameter and the standard deviation parameter of the normal distribution.

Appendix D. Identified weight values and bias values in the ANN model training

See Table D.1.

Table D.1

Identified weight values and bias values in the ANN model training.

Number	Weight value					Bias value
	<i>H</i>	<i>D</i>	<i>M</i>	<i>T</i>	<i>RH</i>	
1	0.17	0.00	0.24	−1.71	−1.24	−1.65
2	−1.02	0.56	−1.28	−0.94	1.13	1.77
3	1.50	−0.51	1.42	0.06	−0.01	−1.09
4	1.30	−0.46	1.80	−0.19	0.44	−1.07
5	1.68	−0.41	1.07	0.76	−0.32	−0.55
6	−1.10	0.91	0.21	−1.00	1.49	0.12
7	1.11	−1.26	1.09	−0.05	1.03	0.02
8	1.30	−1.12	−1.04	0.95	0.43	0.31
9	0.20	0.62	−1.49	0.76	1.43	0.87
10	−0.19	−1.59	−0.21	1.23	−0.89	−1.51
11	−0.85	−0.45	−1.31	−1.59	−0.35	−1.77
12	−2.03	−0.06	−0.34	1.46	−0.34	−1.80

Remark: *H*, *D*, *M*, *T*, and *RH* represent input variables of the ANN model.

Appendix E. Identified parameter values in the probabilistic magnitude model training

See Table E.1.

Table E.1

Parameter values for the fitted probability density functions of the peak abnormal differential load magnitude.

	Normal		Gamma		Weibull	Lognormal		
Parameter	μ	σ	α	β	λ	k	φ	γ
Value	109.8	75.5	2.3	47.7	122.9	1.6	4.5	0.7

Appendix F. Generated hourly probabilistic temperature forecasts in a month using different means of probabilistic weather forecasts

Fig. F.1 shows the 90% prediction interval of the hourly probabilistic temperature forecasts and the actual reported temperatures in October. The boundaries of the 90% prediction interval are the 5% quantile and 95% quantile, respectively.

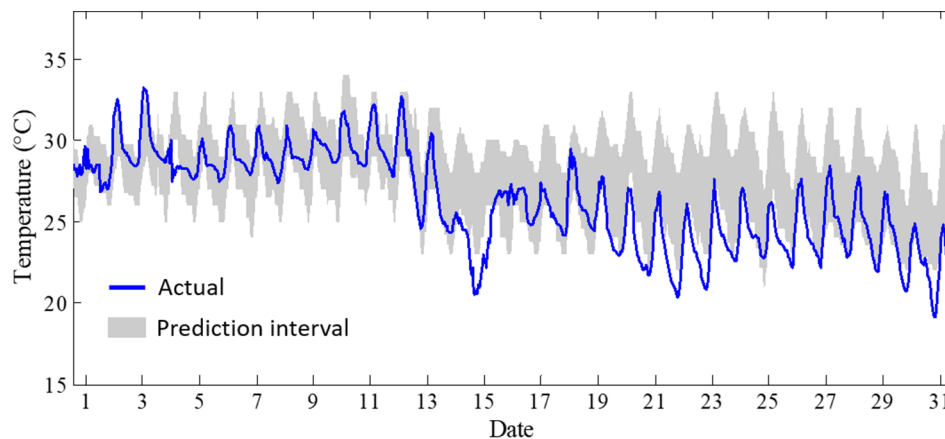


Fig. F.1. Generated hourly probabilistic temperature forecasts of October. The shading represents the boundaries of the 90% prediction interval.

Fig. F.2 shows the 90% prediction interval of the hourly probabilistic temperature forecasts using one-day-ahead weather forecasts and the actual reported temperatures in December. It can be seen that most of the actual temperatures were in the middle of the prediction interval.

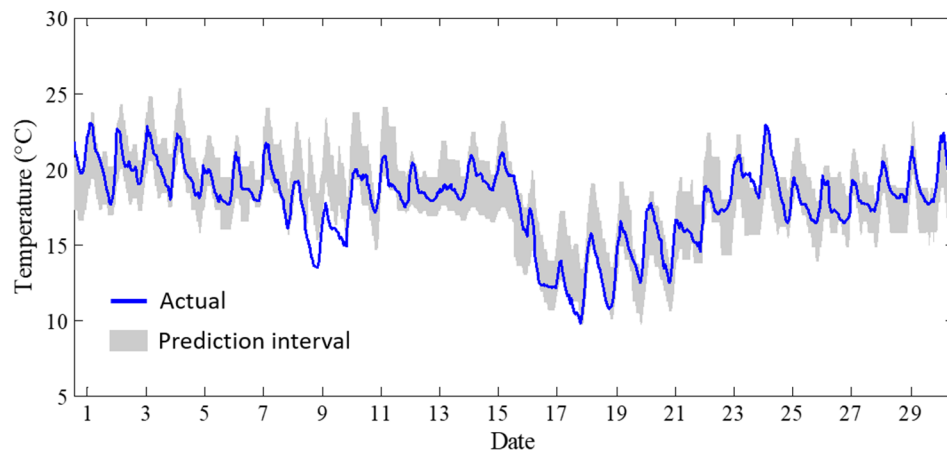


Fig. F.2. Generated hourly probabilistic temperature forecasts of December using one-day-ahead weather forecasts. The shading represents the boundaries of the 90% prediction interval.

Fig. F.3 shows the 90% prediction interval of the hourly probabilistic temperature forecasts using nine-day-ahead weather forecasts and the actual reported temperatures in December. It can be seen that most of the actual temperatures were also in the middle of the prediction interval, but the average bandwidth of the prediction interval was larger than that in Fig. F.2.

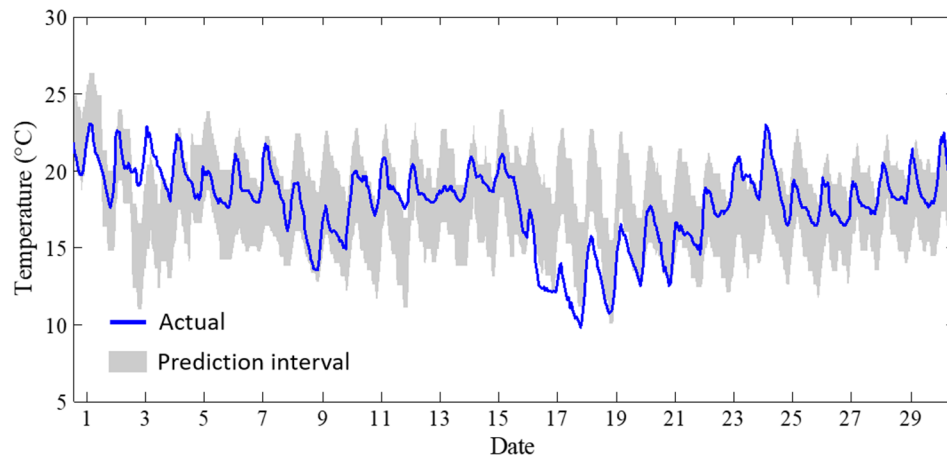


Fig. F.3. Generated hourly probabilistic temperature forecasts of December using nine-day-ahead weather forecasts. The shading represents the boundaries of the 90% prediction interval.

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